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Smart Attendance Management System

A computer vision-based attendance management system that uses **real-time face recognition** to automatically mark student attendance. Built with Python, YOLOv8, OpenCV, and Tkinter.

✨ Features

- **Face Enrollment** — Register students by capturing multiple face photos via webcam
- **Real-Time Recognition** — Automatically identify enrolled students from live camera feed
- **Auto Attendance Logging** — Attendance is recorded to styled Excel (**.xlsx**) files instantly
- **Modern Dark GUI** — Sleek, dark-themed Tkinter interface with status indicators

- **No dlib Required** — Uses YOLOv8-face detector + LBPH recognizer (no C++ compilation needed)
- **Auto Model Download** — YOLO face detection model is downloaded automatically on first run
- **Student Management** — View, enroll, and delete students from the GUI



Tech Stack

Component	Technology
Language	Python 3.8+
Face Detection	YOLOv8-face (ultralytics + yolov8n-face.pt)
Face Recognition	OpenCV LBPH (Local Binary Patterns Histograms)
GUI	Tkinter + ttk (dark theme with custom styling)
Image Processing	OpenCV, NumPy, Pillow
Attendance Logs	openpyxl (styled Excel spreadsheets)



Project Structure

```
attendance_system/
├── main.py                # Main application & GUI (entry point)
├── config.py              # All configuration constants & paths
├── face_detector.py       # YOLOv8-face detection module
├── face_encoder.py        # Student enrollment & LBPH model training
├── face_recognizer.py     # Real-time face recognition module
├── attendance_logger.py   # Excel attendance logging module
├── requirements.txt       # Python dependencies
├── known_faces/           # Enrolled student face photos
│   ├── <rollno_name>/    # Per-student directory
│   │   ├── face_1.jpg    # Cropped grayscale face (200×200)
│   │   ├── face_2.jpg
│   │   └── photo_1.jpg   # Full reference frame
├── encodings/             # Trained model & label data
│   ├── lbph_model.yml    # Trained LBPH recognizer model
│   └── label_map.pkl      # Name ↔ label ID mapping
├── attendance_records/    # Generated attendance Excel files
│   └── attendance_YYYY-MM-DD.xlsx
```

```
└─ models/                                # YOLO model file (auto-downloaded)
   └─ yolov8n-face.pt                     # YOLOv8 face detection model
```



Getting Started

Prerequisites

- **Python 3.8+** installed
- A **webcam** connected to your computer

Installation

1. Clone / navigate to the project directory:

```
cd d:/ML/attendance_system
```

2. Install dependencies:

```
pip install -r requirements.txt
```

This installs:

Package	Version	Purpose
opencv-contrib-python	4.13.0.92	Face recognition (LBPH), image processing
ultralytics	latest	YOLOv8 face detection
numpy	2.4.2	Array operations
openpyxl	3.1.5	Excel file read/write
Pillow	12.1.1	Image display in Tkinter

3. Run the application:

```
python main.py
```

On first run, the YOLO face detection model (~10 MB) will be **downloaded automatically**.



Usage Guide

1. Enroll Students

1. Click  **Enroll Students** in the header
2. Enter the student's **Name** and **Roll Number** in the side panel
3. Face the camera — a yellow bounding box appears when a face is detected
4. Click  **Capture** to take a photo (5 photos required by default)
5. Once all photos are captured, click  **Save Enrollment**
6. The LBPH model is retrained automatically with the new student

2. Take Attendance

1. Click  **Take Attendance** in the header
2. The camera starts and begins recognizing enrolled faces in real-time
3. **Recognized students** are shown with a green bounding box and their name + confidence score
4. **Unknown faces** are shown with a red bounding box
5. Attendance is **automatically marked** when a student is recognized (each student is recorded only once per session)
6. Click  **Stop & Save Attendance** to end the session

3. Manage Students

- The **side panel** shows all enrolled students when no mode is active
- Select a student and click  **Delete Selected** to remove them and their face data

4. View Attendance Records

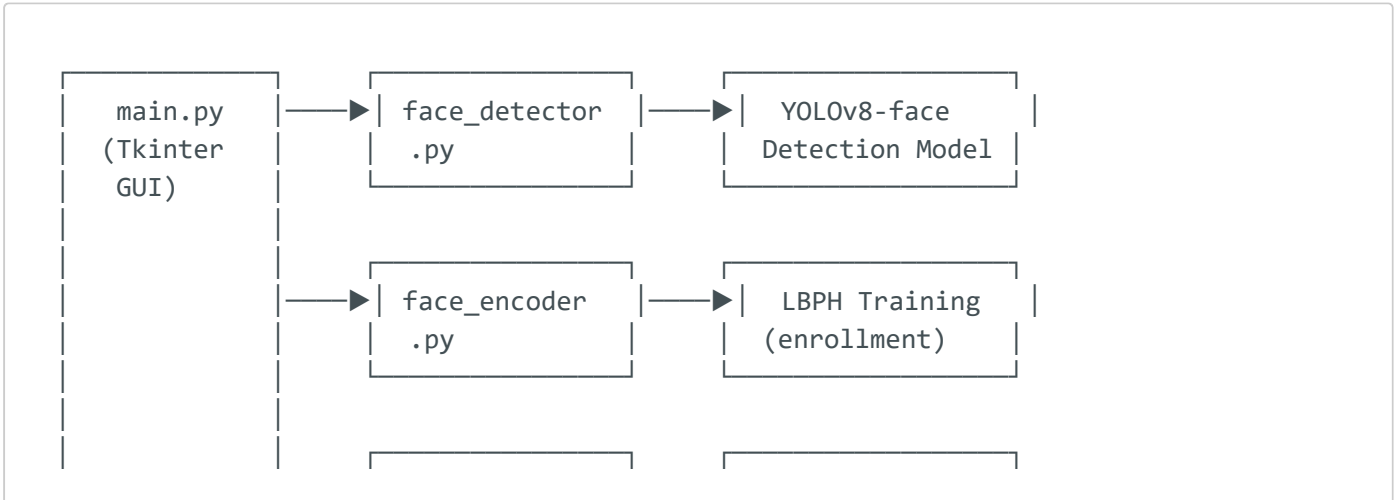
- Attendance files are saved in the `attendance_records/` directory
- Files are named `attendance_YYYY-MM-DD.xlsx`
- Each Excel file contains styled columns: S.No, Roll No, Student Name, Date, Time, Status

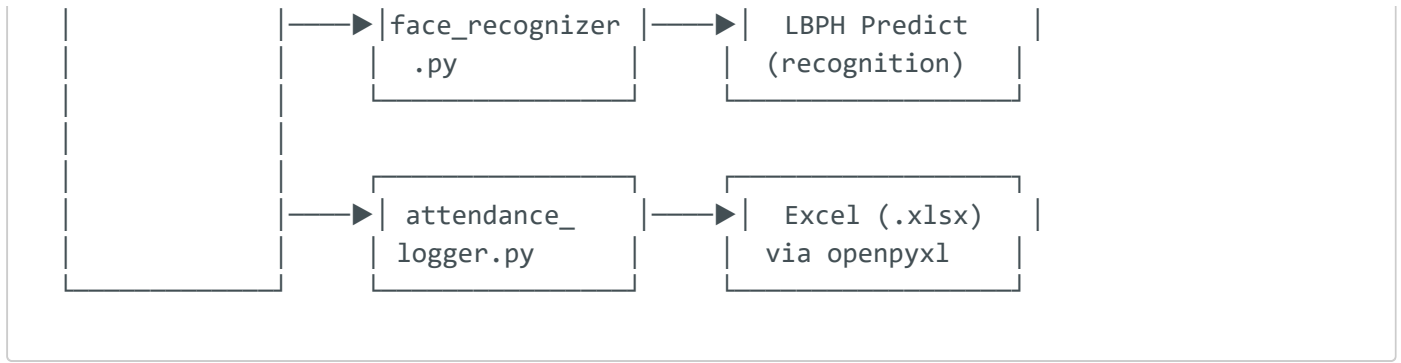
Configuration

All settings are in `config.py`:

Setting	Default	Description
<code>YOLO_CONFIDENCE_THRESHOLD</code>	<code>0.5</code>	Minimum confidence for YOLO face detection (0.0–1.0)
<code>LBPH_THRESHOLD</code>	<code>80</code>	Recognition strictness (lower = stricter, range 50–100)
<code>NUM_ENROLLMENT_PHOTOS</code>	<code>5</code>	Photos captured per student during enrollment
<code>CAMERA_INDEX</code>	<code>0</code>	Webcam index (change if you have multiple cameras)
<code>WINDOW_WIDTH / HEIGHT</code>	<code>1200×750</code>	Application window dimensions

Architecture





Pipeline:

- 1. **Detection** → YOLOv8-face locates faces in each frame
- 2. **Recognition** → LBPH compares detected faces against trained model
- 3. **Logging** → Recognized students are auto-logged to a styled Excel sheet

Excel Output Format

Each attendance file is a styled spreadsheet:

S.No	Roll No	Student Name	Date	Time	Status
1	101	John Doe	2026-03-08	09:15:23	Present
2	102	Jane Smith	2026-03-08	09:15:45	Present

- Header row is styled with dark background and white text
- Data rows have a light green fill
- Columns are auto-sized for readability

Troubleshooting

Issue	Solution
Camera not opening	Check webcam connection, or change <code>CAMERA_INDEX</code> in <code>config.py</code>
Poor recognition accuracy	Capture more enrollment photos in varied lighting/angles

Issue	Solution
Too many false positives	Lower <code>LBPH_THRESHOLD</code> (e.g., 60) in <code>config.py</code>
Too many false negatives	Raise <code>LBPH_THRESHOLD</code> (e.g., 100) in <code>config.py</code>
Model download fails	Manually download <code>yolov8n-face.pt</code> and place it in <code>models/</code>



License

This project is for educational purposes.